Arya Krup | Kiara Jamal | Watson Blair

Predictive Analysis

Predicting total amount of Retail Sales Dataset by applying Machine Learning techniques to build a predictive regressive model.

About The Data

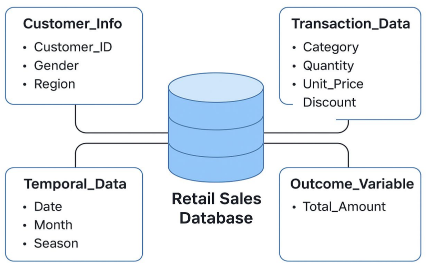
Problem Definition

Feature Engineering

Predictive Modeling

Future Insights

About the Data



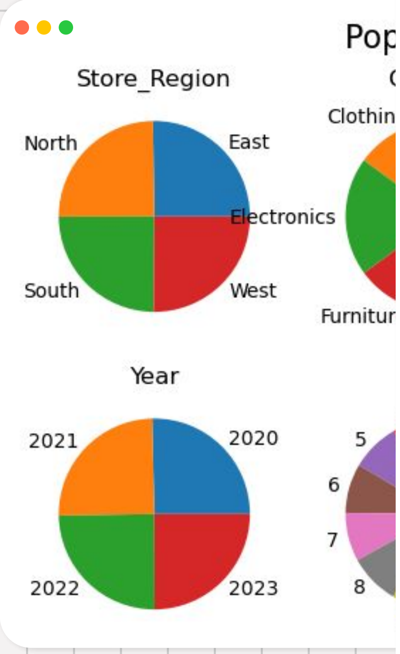
Dataset: Retail Sales Dataset
Size: 50,000 records x 12 columns
Key Features:

- **Customer Info:** Customer_ID, Age, Gender, Store_Region
- **Transaction Data:** Category, Quantity, Unit_Price, Discount, Payment_Method, Online_Or_Offline.
- **Temporal Data:** Date
- **Outcome Variable:** Total_Amount

Use Case

Supports predictive modeling for sales forecasting, customer segmentation and marketing optimization.

Po n



Our data set is almost perfectly balanced across all categorical splits, which indicates that it is not well suited to predictive tasks.

About The Data

Problem Definition

Feature Engineering

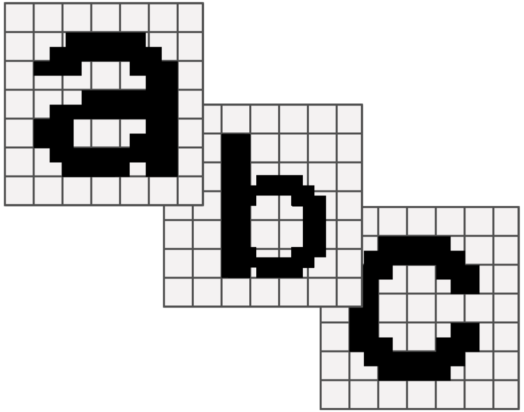
Predictive Modeling

Future Insights

Problem Definition

Outline

Develop a predictive Machine Learning regression model to estimate Total Sales Revenue using various customer, transaction, and temporal features from Retail Sales Data.



About The Data

Problem Definition

Feature Engineering

Predictive Modeling

Future Insights

Feature Engineering

Manipulating data to maximize its utility in modeling tasks

For this data set there were 4 main types of feature engineering required

- Time Feature Destructuring
- Label Encoding
- One-Hot Encoding
- Ordinal Encoding
- Target Encoding

To support different model types and predictive tasks, we created a utility function to generate our engineered features.

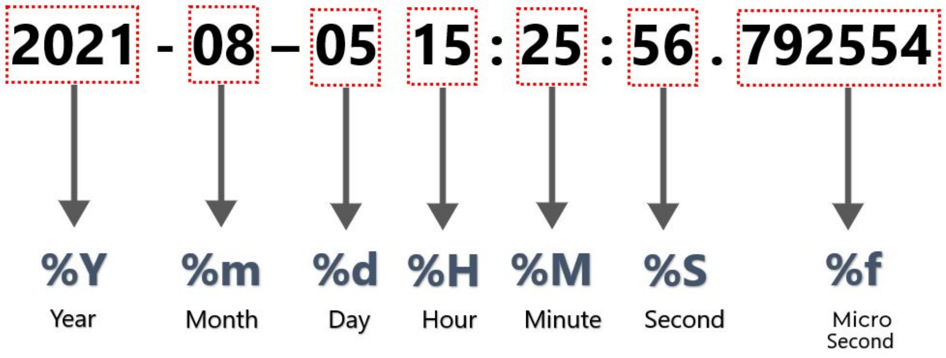
Pro Tip

Time Feature Destructuring

The `Date` feature is a pandas datetime object which we converted into:

- year
- month
- day
- day_of_week

Destructuring the date object like this allows us to filter, sort and group records more easily.



Data Encoding

All data encoding strategies convert categorical features into numerical ones.

Label Encoding
Assigns a unique integer to each categorical value. This is the most basic form of encoding, as it does not account for the ordinal nature of numeric values.

One-Hot Encoding
Creates a binary feature for each categorical value. This removes numeric ordinality from the features, but generates a considerable number of new features.

Ordinal Encoding
Assigns a unique integer to each categorical value as the result of some computation. This ensures that the ordinal nature of numeric values represents a meaningful facit of the data.

Target Encoding
Assigns the mean of the target value when records are grouped by the categorical value.

About The Data

Problem Definition

Feature Engineering

Predictive Modeling

Future Insights

Modeling Methodologies

a. Linear Regression

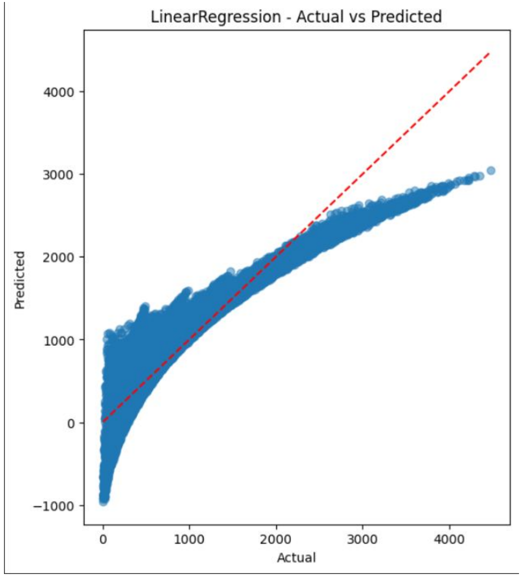
b. Decision Tree

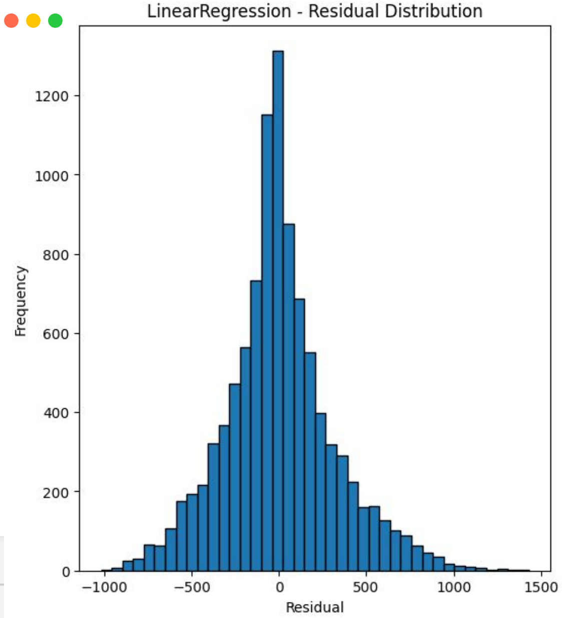
c. Random Forest

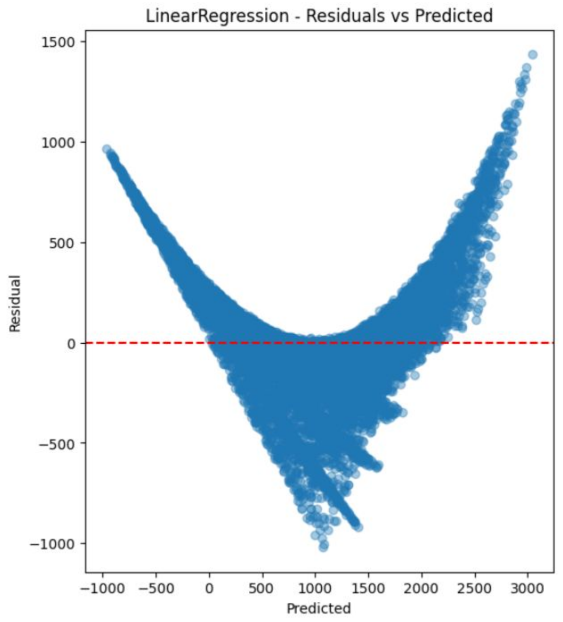
d. XGB Regressor

e. Gradient Boosting

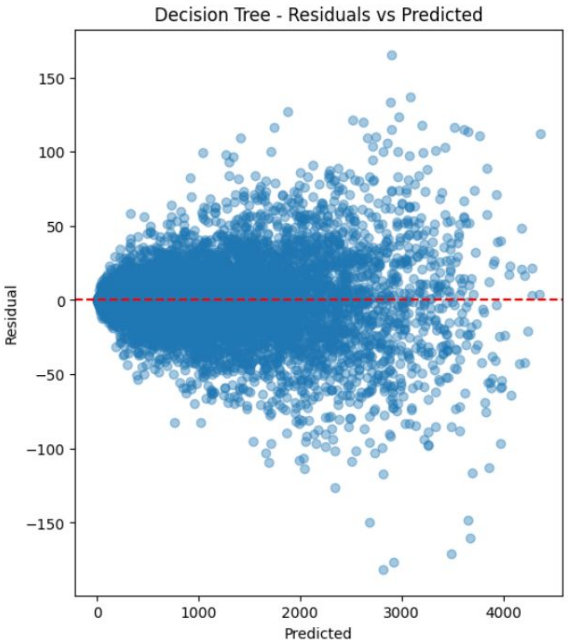
Linear Regression

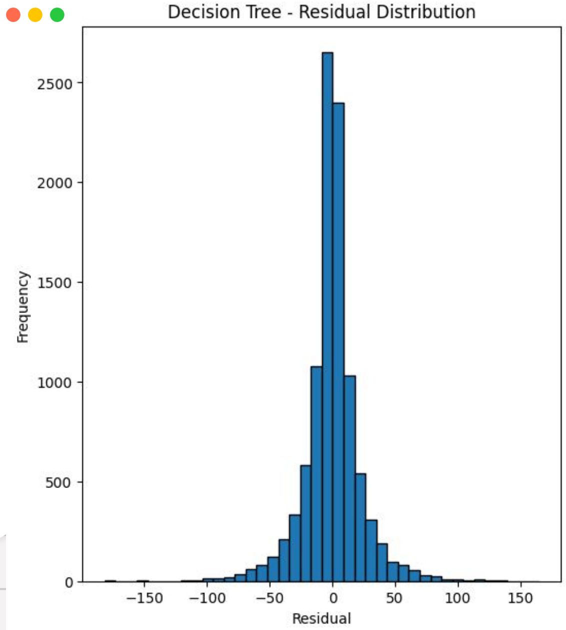


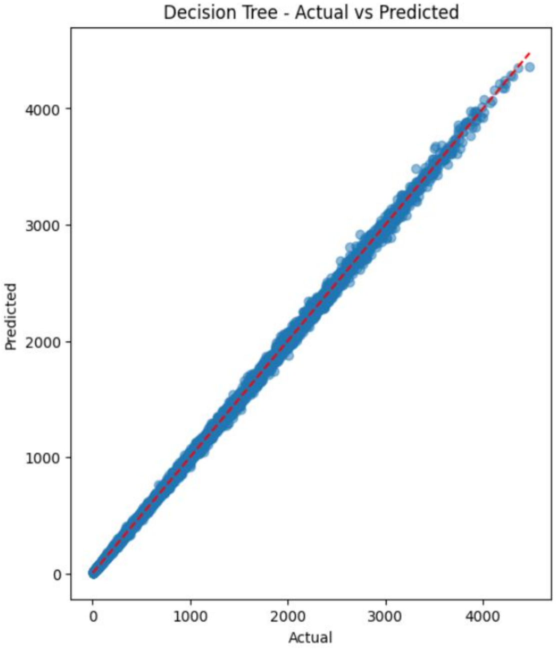




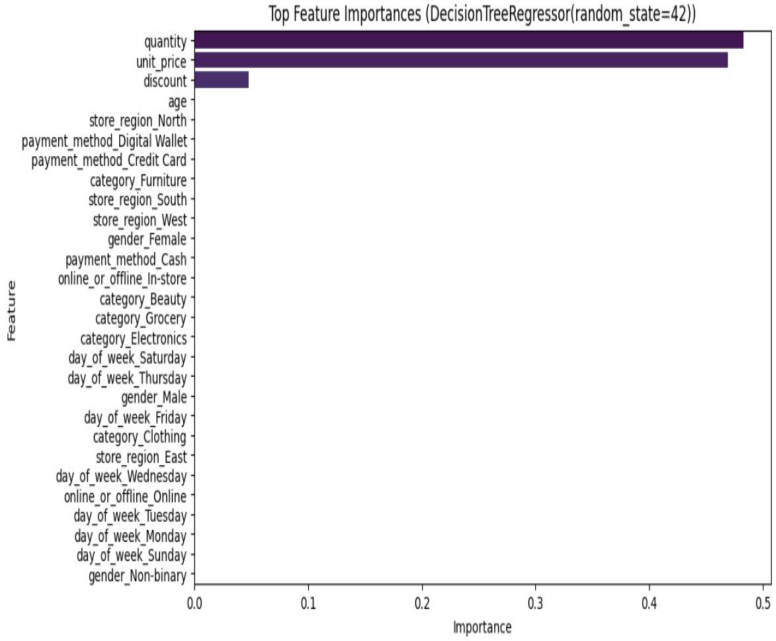
Decision Tree





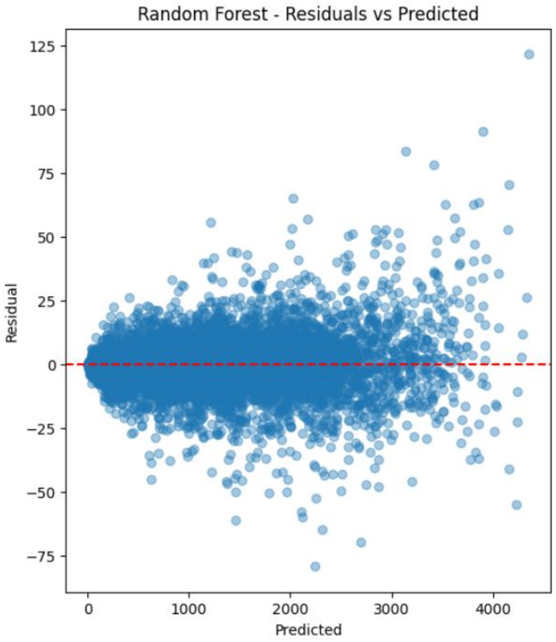


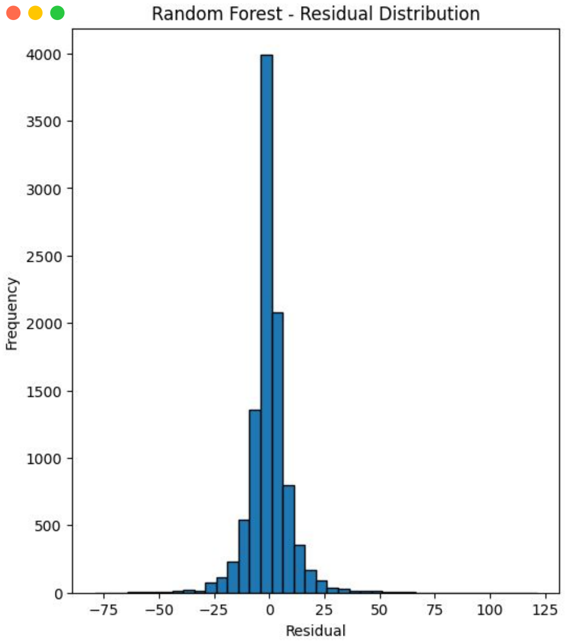
Decision Tree

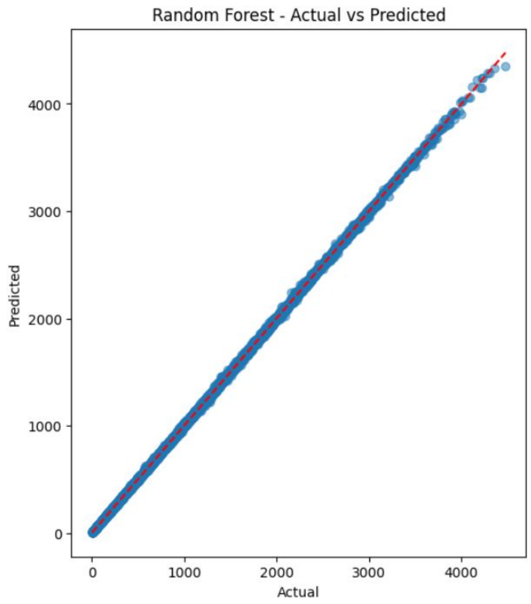


- Tight clustering along the diagonal indicating highly accurate predictions.
- Residuals are evenly distributed around zero with uniform variance.
- Quantity and unit_price are the primary drivers, with discount playing a minor role

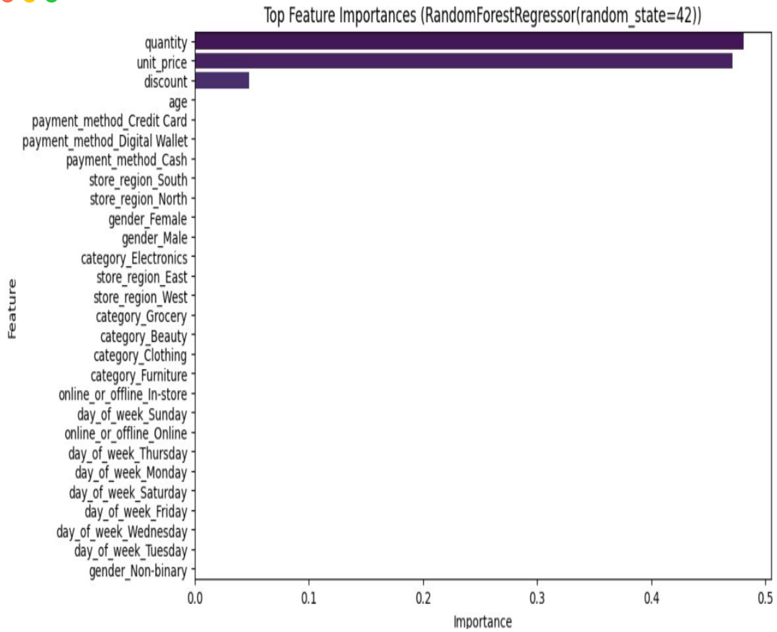
Random Forest





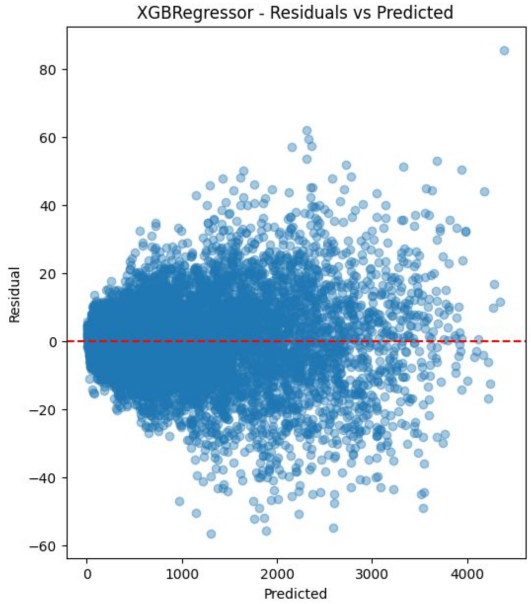


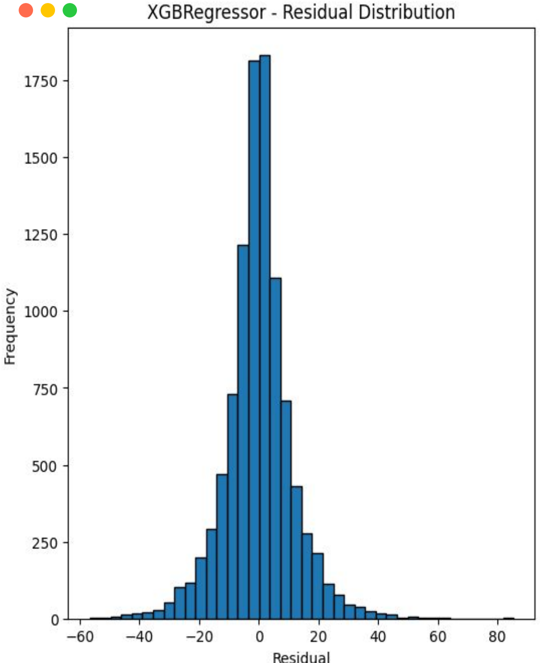
Random Forest

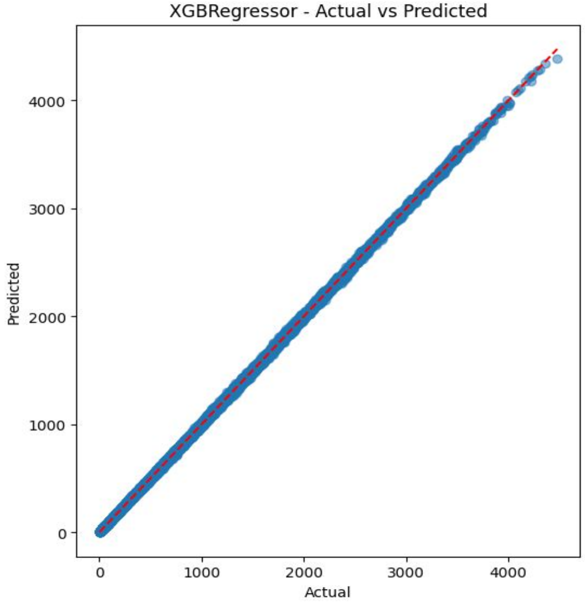


- Great accuracy with predictions tightly aligned along the diagonal, indicating minimal prediction error across all transaction values.
- Residuals are tightly concentrated between -50 and +50, demonstrating significantly improved prediction precision through ensemble learning.
- Quantity (~0.50) and unit_price (~0.48) remain the important features, with discount (~0.12).

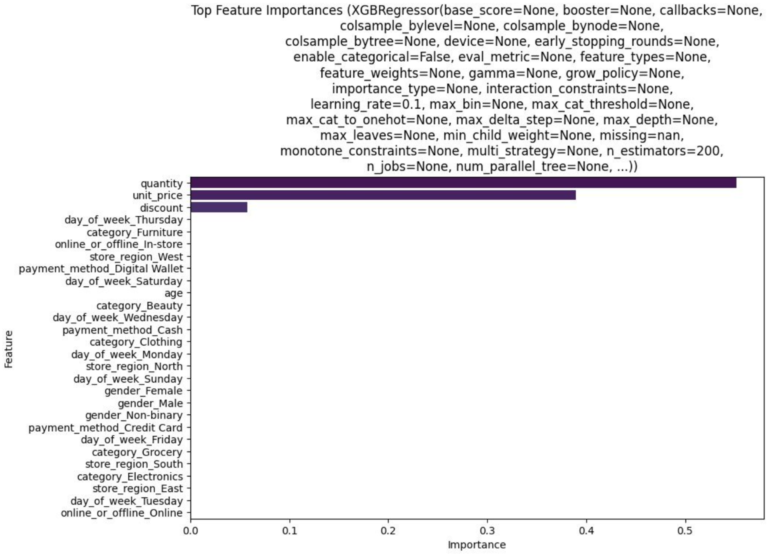
XGB Regressor





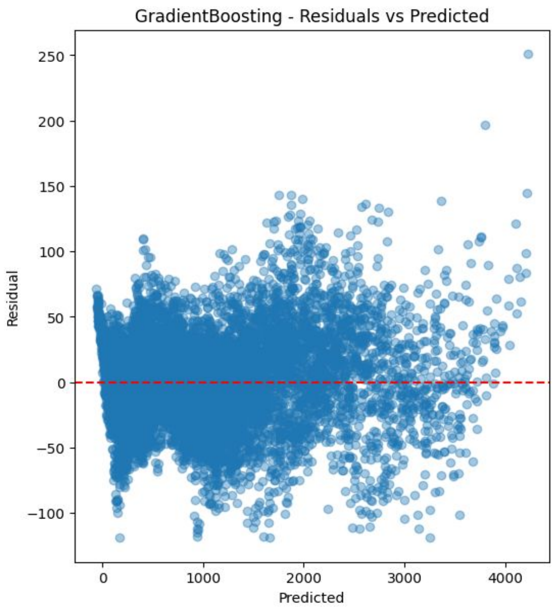


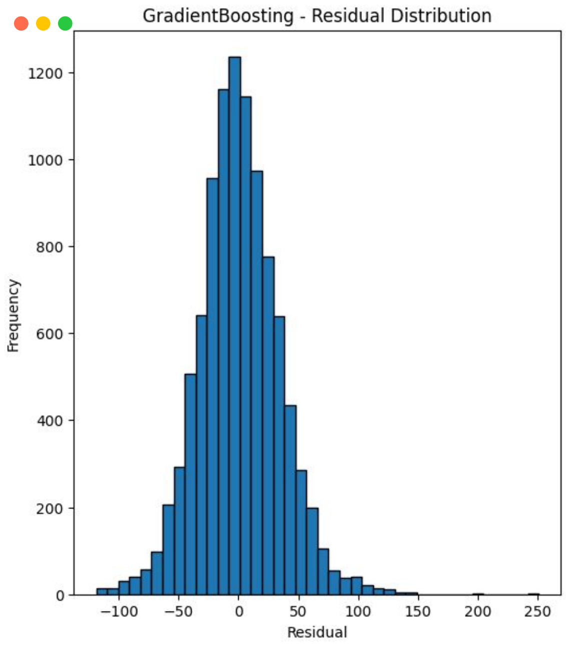
XGB Regressor

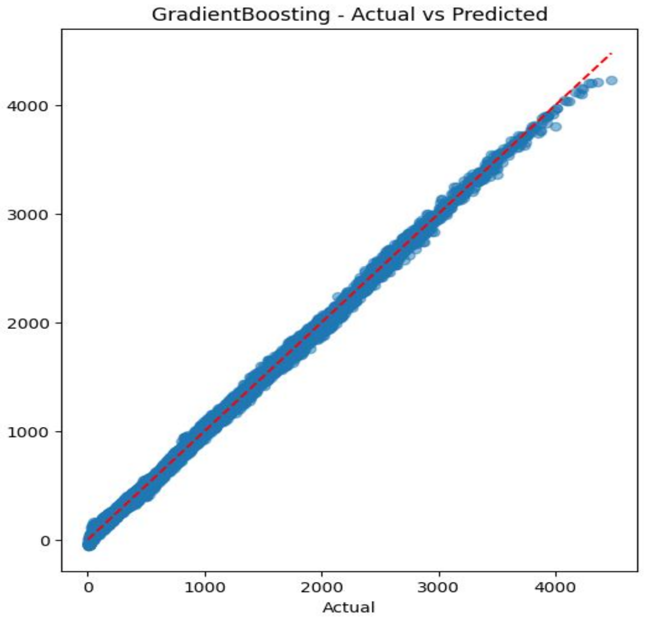


- Achieves the tightest fit along the diagonal line, demonstrating the highest prediction accuracy among all models tested.
- Residuals are compressed to approximately ±60, with most errors concentrated between ±40.
- Quantity (~0.55) and unit_price (~0.40) remain dominant predictors, with discount showing slightly increased importance (~0.08).
-

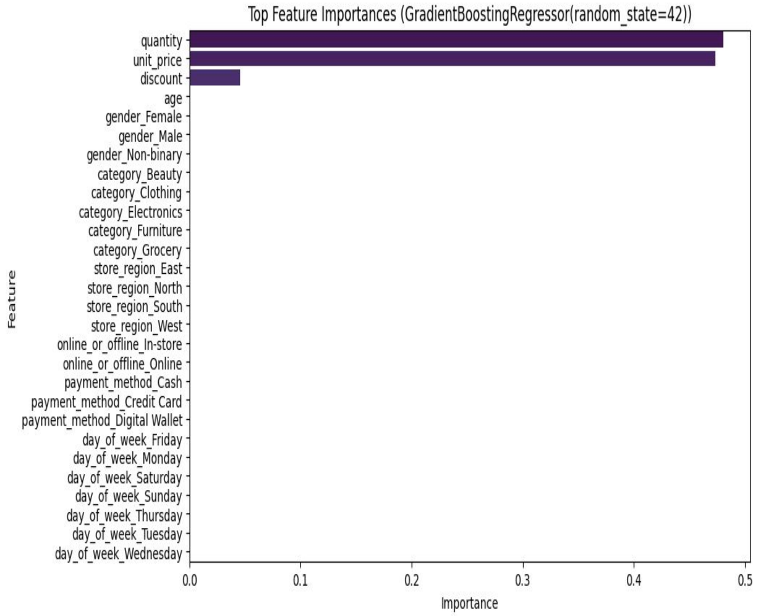
Gradient Boosting







Gradient Boosting



- Strong alignment along the diagonal with slight scatter at higher values which shows good accuracy.
- Residuals span -100 to +250, less consistent prediction precision.
- Quantity (50%) and unit_price (48%) drive predictions, with discount (8%).

Model Performance Summary

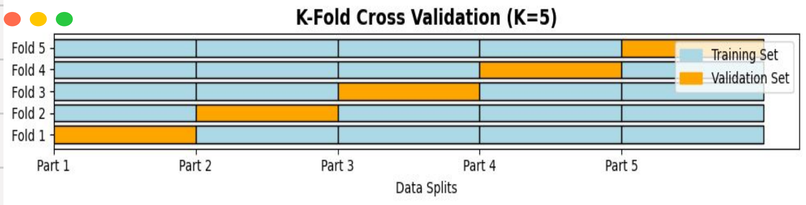
Best Performing Model

	Linear Regression	Decision Tree Regressor	Random Forest Regressor	XGBoost Regressor	Gradient Boosting Regressor
R2	0.859560	0.999267	0.999875	0.999819	0.998397
MAE	232.258172	14.692924	5.868758	7.989741	25.969342
MSE	100041.108	522.168027	89.389412	129.162506	1141.95358
RMSE	316.292757	22.850996	9.454597	11.364968	33.792804

Cross Validation

Model Validation – K-Fold Cross Validation

- Applied 5-fold cross validation, dividing the dataset into 5 parts - training on 4 and validating on 1 iteratively.
- Ensures each observation is used for training & validation, providing a more reliable performance metric.
- Models were evaluated using R^2 Score - higher values indicate better performance.



Purpose

Cross Validation helps evaluate model performance by testing on multiple data splits, ensuring generalization & avoiding overfitting

About The Data

Problem Definition

Feature Engineering

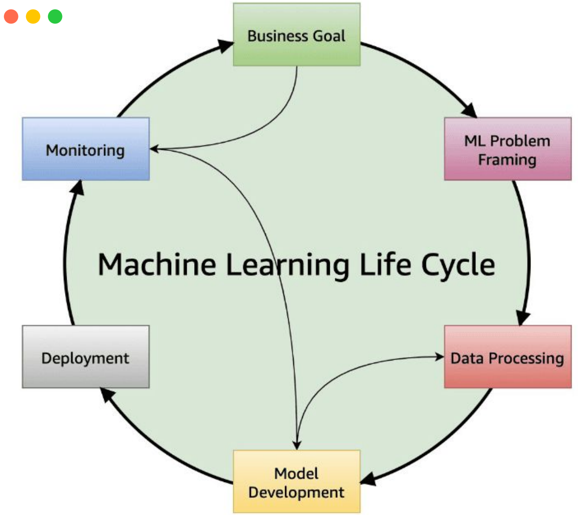
Predictive Modeling

Future Insights

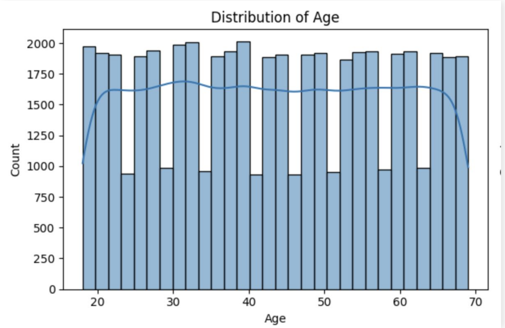
Further Predictions

Forecasting Retail Trends with ML

- This framework shows how ML can forecast sales and reveal customer trends.
- With real data models can generate powerful business insights.
- The following sections explore key predictive opportunities and their business impact.



Total Amount by Demographics

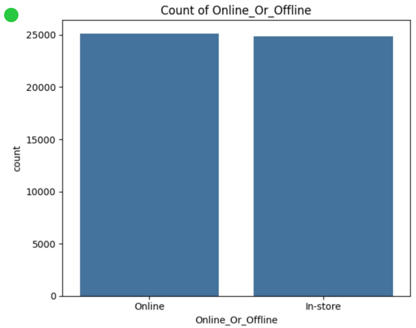


Use **age, gender, income** and **occupation** to estimate spending behavior.

- **Younger customers**: frequent purchases, lower spend per visit.
- **Older customers:** fewer purchases, higher spend.
- **Gender insights** can highlight category preferences

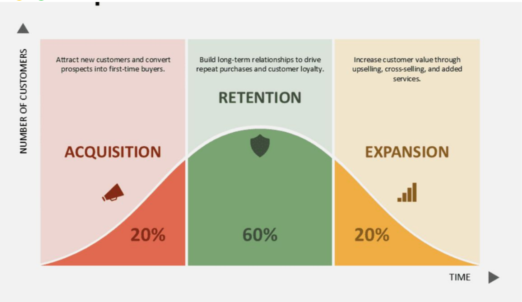
Regional & Store-Level Forecast

- **Demand Forecasting:** Prevent overstock or shortages
- **Regional Promotions:** Tailor offers to local trends.
- **Channel Optimization:** Compare online vs. offline.



Adding geographic data- **city, store, sales channel** to find growth hotspots.

Customer Lifetime Value (CLV)



CLV?

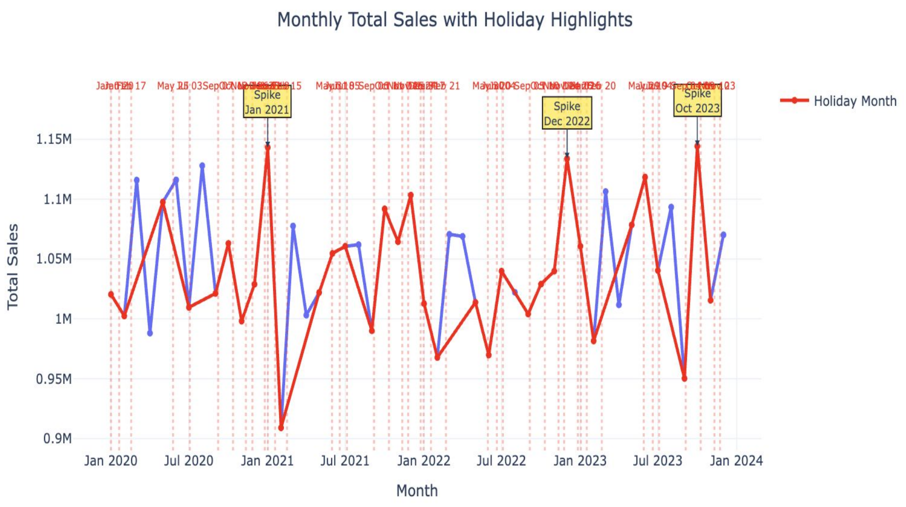
CLV is the total revenue a business can expect to earn from a customer over the years.

- Predict **total revenue** from each customer over time.
- Identify **high-value** and **at-risk customers**
- Guide **loyalty programs** and **personalized offers**
- Forecast overall revenue from aggregated CLV.

Product Category & Purchase Patterns

- Analyze which **categories sell most** across demographics and regions.
- Understand how **discounts influence sales.**
- Detect products often **bought together** for **bundling** & **stock control**.

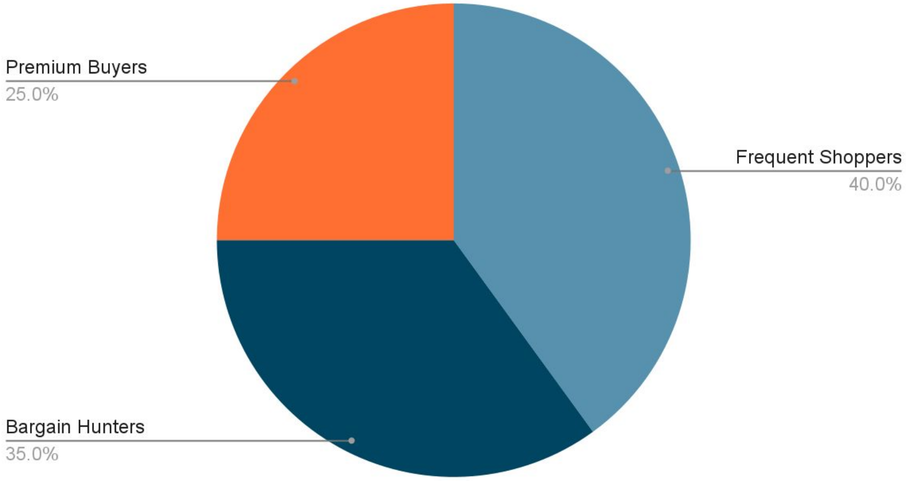
Temporal & Seasonal Spending



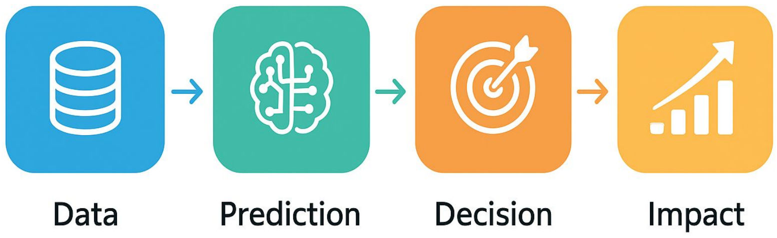
- Detect **time-based** patterns by **day, month** or **season**.
- Like **holiday peaks** or **weekend** shopping trends.
- Use models like **ARIMA or Prophet** to forecast future sale.
- Combine **time & demographic data** for deeper insights.

Behavioral Segmentation & Personalization

- Apply clustering to **group customers** by spending habits.
- Identify **frequent shoppers**, **bargain hunters**, **premium buyers**.
- Personalize **recommendations** and **marketing campaigns**.



Business Impact



- Identify high-value segments and profitable products.
- Optimize marketing spend across customer groups.
- Reveal top-performing regions and seasons.

Conclusions

- Built a complete ML regression pipeline - from feature engineering to evaluation.
- Synthetic dataset limited accuracy, but validated a scalable predictive framework.
- Demonstrated Applications for:
  – Sales Forecasting by Demographics & Region
  – Customer Lifetime Value (CLV) estimation
  – Behavioral Segmentation & Personalization
- Framework Ready for real-world retail data integration and business insights
- Next steps: Automate hyperparameter tuning, extend to time-series forecasting.

Thank You

Questions?